The code is called from IDL as

```
res = call_external(libname, 'GET_MW', Lparms, Rparms, Sparms, Bparms,  $
                    Flux)
```

where libname is the library name (*.dll or *.so), and the parameters are:

0. Lparms – 4-element long integer (32-bit) array of dimensions and global integer flags.
1. Rparms – 3-element floating point (double, 64-bit) array of global real parameters.
2. Sparms – 5-element floating point (double, 64-bit) array of the source parameters.
3. Bparms – two-dimensional (14 × Ncomp) floating point (double, 64-bit) array describing the particle distributions.
4. Flux – two-dimensional (3 × Nf) floating point (double, 64-bit) array receiving the computation results.

Namely, the above arrays have the following components:

Lparms = [Nf, Ncomp, Nnodes, FFflag]

where

0. Lparms[0] = Nf – number of frequencies in the spectrum.
1. Lparms[1] = Ncomp – number of considered plasma components.
2. Lparms[2] = Nnodes – number of energy nodes used for integration over energy in the continuous gyrosynchrotron code. Minimum value: 16; if Nnodes < 16, 16 notes will be used instead.
3. Lparms[3] = FFflag – controls the free-free contribution: if zero, the free-free contribution (determined by the thermal plasma parameters $n_{\rm th}$ and T_0, and computed using the approximate formulae from Dulk 1985) is included; if nonzero, the free-free contribution is ignored.

Rparms = [S, f_0, Δf]

where

0. Rparms[0] = S – visible source area, in cm^2.
1. Rparms[1] = f_0 – starting frequency of the spectrum, in Hz. If $f_0 > 0$, the emission frequencies are given by f_0, $f_0 \times 10^{\Delta f}$, $f_0 \times 10^{2\Delta f}$, $f_0 \times 10^{3\Delta f}$, ... If $f_0 \leq 0$, the emission frequencies are taken from the Flux[0, *] array.
2. Rparms[2] = Δf – logarithmic frequency step used to produce the spectrum (is used only when $f_0 > 0$).

Sparms = [Δz, $n_{\rm th}$, T_0, B, θ]

where

0. Sparms[0] = Δz – source depth along line-of-sight, in cm.
1. Sparms[1] = $n_{\rm th}$ – thermal electron concentration, in cm^{-3}.
2. Sparms[2] = T_0 – thermal electron temperature, in K.

3. Sparms[3] = B – magnetic field strength, in G.
4. Sparms[4] = θ – viewing angle, in degrees.

Bparms = dblarr(14, Ncomp), where Ncomp is the number of the considered plasma components, and the parameters of the i-th component are specified as

0. Bparms[0, i] = q – particle charge, in units of the electron charge. E.g., q =1 for electrons and muons, q = -1 for positrons and protons, q = -2 for α-particles.
1. Bparms[1, i] = m – particle mass, in units of the electron mass. E.g., m =1 for electrons and positrons, $m \approx 1836$ for protons.
2. Bparms[2, i] = n – concentration of the considered particles, in cm^{-3}.
3. Bparms[3, i] = Eid – specifies the chosen energy distribution (non-integer parameters are rounded to the nearest integer).
4. Bparms[4, i] = either E_{min} or T:
   a) E_{min} – the minimum particle energy for the power-law and double power-law distributions, in MeV;
   b) T – the temperature for the thermal distribution, in K.
5. Bparms[5, i] = E_{max} – the maximum particle energy, in MeV.
6. Bparms[6, i] = E_{break} – the break energy in the double-power-law particle distributions, in MeV.
7. Bparms[7, i] = δ_1 – the low-energy power-law index.
8. Bparms[8, i] = δ_2 – the high-energy power-law index in the double-power-law particle distributions.
9. Bparms[9, i] = μid – specifies the chosen pitch-angle distribution (non-integer parameters are rounded to the nearest integer).
10. Bparms[10, i] = α_c or α_0 – either the loss-cone boundary in the loss-cone distributions or the beam direction in the beam-like distributions, in degrees.
11. Bparms[11, i] = $\Delta\mu$ – either the loss-cone boundary width or the beam angular width in the loss-cone or beam-like distributions, respectively.
12. Bparms[12, i] – *reserved*.
13. Bparms[13, i] = Nflag – specifies whether the considered particles make a contribution into the plasma dispersion: if nonzero, the concentration n = Bparms[2, i] will be added to the thermal electron concentration n_{th} = Sparms[1] before computing the plasma frequency (but will not affect the free-free emission); if zero, the particles are assumed to have no contribution into the dispersion.

Note that Ncomp can be zero; in such case, only the free-free emission is computed.

Flux = dblarr(3, Nf), where Nf is the number of the emission frequencies. After calling the function, this array contains the computation results (and can specify the emission frequencies on input), namely:

0. Flux[0, *] – emission frequencies, in GHz. On input, if Rparms[1] = $f_0 \leq 0$, the frequencies are taken from this array; otherwise, the frequencies are computed automatically.
1. Flux[1, *] – X-mode emission intensities, in sfu.
2. Flux[2, *] – O-mode emission intensities, in sfu.

Returned value: 0 if there were no errors; -1 if there was an insufficient number of parameters (< 5).

Currently, the following energy distributions are implemented:

- Thermal distribution (THM, Eid =2).
- Power-law over the particle energy (PLW, Eid = 3).
- Double power-law over the particle energy (DPL, Eid = 4).
- Power-law over the particle flux (PLF, Eid = 13).
- Double power-law over the particle flux (DPF, Eid = 14).

The following pitch-angle distributions are implemented:

- Isotropic (ISO, μid = 0 or 1).
- Symmetric gaussian loss-cone (GAU, μid = 3).
- Gaussian beam (GAB, μid = 4).

If either Eid or μid do not match the above values, no emission is computed for the corresponding particle component (although the concentration may be added to the thermal electron concentration, if Nflag ≠ 0).